School of Computing

UNIVERSITY OF PORTSMOUTH

Module Title and Code	ADVANCED ARTIFICIAL INTELLIGENCE - M33176 - FHEQ 7
Module Coordinator Other lecturers	Assoc. Prof. Alex Gegov <alexander.gegov@port.ac.uk> Prof. Ivan Jordanov <ivan.jordanov@port.ac.uk>
Assessment Item number	Item 1
Assessment Title	Coursework_1 (AAI - Neural Networks and Genetic Algorithms)
Date Issued	25/02/2026

Schedule and Deliverables

Deliverable	Value	Format	Open Date	Due date
Coursework report	50%	A single **.pdf** file (coursework report) to Moodle.	**14/07/2026** 13:00 [GMT]	**30/07/2026** 13:00

Notes and Advice

- The Extenuating Circumstances procedure is there to support you if you have had any circumstances (problems) that have been serious or significant enough to prevent you from attending, completing or submitting an assessment on time. If you complete an Extenuating Circumstances Form (ECF) for this assessment, it is important that you use the correct module code, item number and deadline (not the late deadline) given above.
- ASDAC are available to any students who disclose a disability or require additional support for their academic studies with a good set of resources on the ASDAC moodle site
- The University takes any form of academic misconduct (such as plagiarism or cheating) seriously, so please make sure your work is your own. Please ensure you adhere to our *Student Conduct Policy* and watch the video on Plagiarism.
- Any material included in your coursework should be fully cited and referenced in APA 7 format. Detailed advice on referencing is available from the library, also see TECFAC 08 Plagiarism.
- Any material submitted that does not meet format or submission guidelines, or falls outside of the submission deadline could be subject to a cap on your overall result or disqualification entirely.
- If you need additional assistance, you can ask your personal tutor, student engagement officer ana.baker@port.ac.uk, academic tutor eleni.noussi@port.ac.uk or your lecturers.
- If you are concerned about your mental well-being, please contact our Well-being service

M33176, *Advanced Artificial Intelligence (Part 1)*
(***Neural Networks and Genetic Algorithms***)

Coursework Assignment 1

This is an assessed piece of coursework (worth 50% of your final mark). If you are unclear about any aspect of the assignment, including the assessment criteria, please raise this at first opportunity. The submitted work should use Matlab Toolboxes and must be your own work. If it includes other people ideas and materials, they must be properly referenced or acknowledged. Failing to do so intentionally or unintentionally constitutes plagiarism. The University treats plagiarism as a serious offence.

Your Task

This coursework worth 50% of your final mark and includes two parts: Applying Neural Networks (NN) for a classification task and Genetic Algorithms (GA) for solving a simple global optimization problem.

The first part of the task is to analyse, design, train, validate and test artificial NN (using Matlab Neural Network Toolboxes), that can solve classification/prediction problem for a given dataset.

You have to develop Neural Network based Machine Learning Model for Classification and Prediction of Osteoporotic Spinal Fractures using the provided dataset (see **Appendix A**). The dataset comprises 600 records of 25 characteristics of patients (plus the label denoting the two classes: *Fracture* (VBX 1); and *Healthy* (VBX 0).

The purpose of this part of the assignment is to learn and demonstrate that you can: analyse the dataset, pre-process (including dealing with noise, missingness, and data imbalance) and visualise it; design neural network topology and architecture; use training algorithm (Backpropagation); validate and test your NN ability to generalise, and post-process the results. You should use Matlab NN toolboxes to design, train neural networks that can classify, predict, and map input samples with expected outputs and subsequently test the NN using appropriate performance metrics (confusion matrices, ROC curves, AUC, F1 score, etc.).

The second part of the coursework is about using GA for solving Global Optimization problem. You will have to use Matlab GA solver to find the global minima of the provided *Shekel* family of multimodal, multidimensional mathematical functions (see **Appendix B**). You should choose appropriate settings for the GA initial population, fitness function, selection technique, reproductive operators, stopping criteria, etc. You will also have to set up adequate parameters (initial population number of chromosomes, stochastic parameters (e.g., probability of mutation), bounds and constraints for some of the parameters, etc.).

Use the Matlab *Live Editor Optimize Task* and the *GA Solver* to solve the problem and to visualize the evolution of the chosen population, the best, the mean, and the standard deviation values of the fitness function at each generation and to report, analyse, and discuss the results. You can also implement different selection strategies (e.g., fitness-proportional, rank-based, tournament selection, etc.), and compare and critically discuss the results. Some discussion should be given on premature convergence, how can it be avoided and how the selection pressure can be kept relatively constant throughout the whole run. In your search strategy pay attention to the characteristics/parameters that can improve the balance between the exploration and exploitation abilities of your global search technique.

Deliverable

By **1pm, 30.VII.2026,** you should submit (to Moodle) a **.pdf** file with your report. The main body of the report should be **maximum of six pages** (up to **3000 words**), excluding the title page and the Appendixes (for which you have no limit). It should be well structured and soundly written, demonstrating critical thinking and comparative analysis of the adopted approaches and the interpretation of the results. Typical content (but not limited to):

1. Abstract;
2. Introduction;
3. NN part:
   - Data Set - Analysis and Pre-Processing;
   - NN analysis and design;
   - Learning/Training analysis and results;
   - Test/Post-Processing/Generalisation results – comparison and analysis;
   - Conclusion and recommendations.
4. GA part:
   - Brief analysis of the chosen function to optimise;
   - Analysis and discussion of the adopted selection strategy, fitness function, and reproductive operators;
   - Analysis and brief discussion of the chosen parameters, stopping criteria, and convergence of the search;
   - Comparative and critical analysis of the results;
   - Conclusion and recommendations.
5. Reference list;
6. Appendices (any results, screen shots, diagrams and figures, and your Matlab code related to the coursework).

Assessing your assignment, I shall be looking for the following:

- Some thought and analysis of what is the complexity of designing, training, testing, validating and using a neural network for solving the problem;
- You have thought about coding the output and appropriate NN topology and architecture, adoption of adequate training technique (appropriate parameters, activation functions, etc.);
- You have thought about what data to use and for what purpose (subsets for training, testing, and validation), and how the subsets are organized (proportions of the dataset, internal distribution, etc.);
- You have thought about what kind of selection strategy to adopt for the GA optimization and what kind of operators to use for the reproduction (e.g., standard crossover and mutation, choice of parameters, e.g., probability of mutation, population number, population gap, etc.);
- Premature convergence and the ways of avoiding it are considered, choice of stopping conditions and strategy to keep the balance between exploration and exploitation of your global search, and also choice of parameters that could keep the selection pressure relatively constant throughout the whole run;
- Finally, you should demonstrate ability to critically analyse, discuss, and interpret the implemented approaches and obtained results, and to give some recommendations for further improvements.

Assessment Criteria

A - An excellent, well-written and structured report that makes an interesting and easy to follow read. NN part - you have the data set explored and analysed in a number of ways (e.g., pre-processing, normalisation, standardisation; how are the different subsets split; will a swap of the training/validation and testing subsets produce different results); appropriate coding, NN topology and architecture are chosen; the training, validation and testing are well analysed and discussed, and you have neural networks that perform well and can generalise. GA part – you have analysed the function in hand and have adopted suitable selection strategy and reproductive operators. The choice of particular parameters and conditions is critically analysed and discussed.
You have analysed the complexity of the problems and the performance of the applied methods (different approaches may have been applied and compared), and have been critically analysed, interpreted, and presented (and visualised) the results in an interesting and sound way.

B - A well-written report that is well structured into an interesting read. You have neural networks that perform well on the data set and you have analysed their performance. Your GA performance is critically analysed and discussed. You have exhibited some initiative in the approach taken and the results are clearly presented and critically discussed and compared.

C - A reasonable report that presents an account of the adopted approaches with critical analysis and discussion at places. The results are presented relatively well and clear.

D - A report that presents the results mostly in descriptive way, but show acceptable understanding of the applied approaches.

F - Either no report submitted, or a report that presents very few results and little or no understanding of how these methods work.

Mark Scheme

Analysis and Pre-Processing (dealing with missing data, normalisation, standardisation) of the datasets and the multimodal function to be optimized.	**10**
NN and GA assumptions, Analysis and Design.	**20**
NN Training, Testing, Post-Processing, Generalisation, and Results. Adopted training, validation and testing strategy analysed and explained – e.g., why ‘split-sample’ or ‘cross-validation’ adopted?	**20**
GA analysis of the Run, Adopted Selection strategy, Reproductive operators and Results.	**20**
Report – structure, style, and presentation. Critical and comparative analysis and discussion demonstrating in-depth understanding and grasp of the investigated methods and approaches. Good visualisation and sound discussion and interpretation of the results.	**30**